

EMMANUEL SAMPSON

ASSOCIATE ELECTRICAL ENGINEER

CONTACT

PHONE

(804) 926-0822

WEBSITE

<https://emmanuel-sam.github.io/portfolio/>

LINKEDIN

[linkedin.com/in/emmanuel-sampson](https://www.linkedin.com/in/emmanuel-sampson)

EMAIL

ecsampson03@gmail.com

Dear Hiring Team,

I am writing to express my interest in Intramotev's Associate Electrical Engineer position. I am a strong candidate for this role due to my passion for engineering, strong subject matter expertise in electrical systems and AI-driven design, and proven communication skills in collaborative technical environments. Throughout my academic journey, I have focused myself towards building a strong foundation in programming, PCB & circuit design (Multisim, Kicad, Autodesk Eagle), CAD Modeling with Autodesk, and in my ability to best relay that information to others through hands-on projects like developing bomb-detecting robots, making hackathon multimedia radio frequency detection programs, and creating convolutional neural networks using Python, C++, and High-Performance Computing (HPC) clusters.

At William & Mary, I double-majored in Engineering Physics and Philosophy, and by doing this, I developed the ability to think critically and analytically, solving problems that required creative and innovative solutions. This expertise has enabled me to tackle complex challenges, such as designing AI-powered computer vision systems for defect detection in fiber optics during my internship at WEINERT Fiber Optics, where I used Python, C++, and HPC clusters to optimize real-time analysis. One such problem was the revival of the William & Mary Robotics Club. Prior to my year, there were no members of the club, but with my technical interests, I made efforts to improve the club's standing with the university despite not knowing much about the topic at the time. I gathered a team, and through our efforts of creating workshops, and hosting events such as the William & Mary Hackathon, I was able to make an organizational system that will live well beyond my college years, enabling students from any background or skill level to build projects like LED Jack-O'-Lanterns, robotic dogs, and award-winning hackathon entries such as the Bird Sensor, which won first place at the William & Mary 2024 Hackathon by detecting avian electromagnetic signatures.

In addition to this, the Engineering Physics major at William & Mary is heavily geared towards both electrical engineering and modeling for mechanical design. We're required to take two electronics courses, a beginner's and an advanced course, with weekly lab projects that include creating remote temperature sensors, NEMA 17-controlled linear actuators, and real-time audio synthesizers. I'd like to describe the many projects I completed in these classes, during hackathons, and in the robotics club, all for the sake of remembering how much fun they were to me. I'm very passionate about electronics, making, and building. This passion drives me to explore beyond requirements, as seen in my makerspace role, where I managed a \$4,000 budget to procure equipment and engineered custom data-acquisition devices for reflective optics experiments. For our final electronics assignment, which involved creating an audio-visual synthesizer that takes voltage input directly from sound, I designed 20 different circuits using Multisim to explore every possible approach. These projects excite me, fueling my commitment to engineering innovation, and I enjoy sharing that excitement with the robotics team I've developed, the over 200 students I've trained at makerspaces on tools like Fusion 360 and laser cutters, and my advisors when completing internship projects, such as polarization alignment for PM fibers.

By joining Intramotev, I aim to further this learning journey to apply my skills in circuit design, mechanical engineering, and team building. I see autonomous rail technology as a project that I can commit to and genuinely enjoy. I have already made arrangements to move closer to the work location if selected. I am eager to utilize the skills I now possess to contribute to innovative rail solutions and to learn more to share.

Thank you for your consideration, and I look forward to discussing how I can contribute to Intramotev.

Sincerely,
Emmanuel Sampson